

Evaluation of imputation methods on missing data on categorical variables

1. Introduction

Missing data are inevitable in research and are particularly common in observational and epidemiologic studies. When a substantial proportion of data is missing, standard complete-case analyses (CCA) can lead to biased estimates, loss of efficiency, and misleading inference.

Blankers et al. (2010) illustrated the impact of different missing data methods using an eHealth dataset with a continuous outcome, where 50% missingness was induced under a missing at random mechanism. Comparing CCA, single imputation methods, expectation–maximization (EM), and multiple imputation by chained equations (MICE), they showed that CCA and approaches produced biased estimates, whereas multiple imputation methods yielded estimate with small bias and good coverage of the true parameters.

Motivated by this literature, the aim of this report is to investigate the performance of different missing-data strategies in the context of an ordinal outcome model. A proportional odds model for the number of affairs in the past year was fitted using the Affairs dataset from the Applied Econometrics with R package. Missingness at three different levels was then introduced according to a pre-specified mechanism and subsequently handled using four approaches: CCA, EM-based single imputation (SIEM), EM-based multiple imputation (EM), and MICE. Their performance was evaluated in a simulation study in terms of bias and root mean squared (RMSE) error. Overall, the results indicate that SIEM consistently outperforms the other methods across these metrics, while CCA exhibits the poorest performance.

The remainder of the report is organized as follows. Section 2 presents the data and the model used. Section 3 describes the missingness mechanism and the missing data approaches. Section 4 details the structure of the simulation study. Section 5 presents the results and Section 6 summarizes the conclusions.

2. Data and Model

2.1. Dataset

The dataset used is Affairs from the AER package. This dataset includes 601 observations and 9 variables: number of affairs in the past year, age, gender, years of marriage, presence of children, religiousness, education, occupation, and self-rating marital satisfaction.

All variables are coded categorically, with gender and children being binary and the remaining variables ordinal. Due to sparsity issues in the original categories, the number of affairs was recoded into three groups, “0 times,” “1–3 times,” and “more than 3 times”, and treated as an ordinal outcome variable for the analysis. A particular interest lies in exploring the association between the number of affairs and the presence of children, gender, religiousness, and self-rated marriage satisfaction.

2.2 Model

2.2.1 Proportional odds model

Let Y denote an ordinal response with K ordered categories, labeled $1, \dots, K$ and X be a vector of covariates. The proportional odds model specifies a separate logistic regression for each cumulative probability $\Pr(Y \leq k|X)$, but with a common slope vector across cutpoints, $k = 1, \dots, K - 1$ with ordered intercept $\alpha_1 \leq \alpha_2, \dots \leq \alpha_{K-1}$.

$$\text{logit}\{\Pr(Y \leq k|X)\} = \log \left[\frac{\Pr(Y \leq k|X)}{\Pr(Y > k|X)} \right] = \alpha_k - \beta^T X \quad (1),$$

where α_k are category specific intercepts and β is a common vector of regression coefficients.

The cumulative odds ratio is obtained by:

$$\text{odds}(Y \leq k|X) = \exp(\alpha_k - \beta^T X).$$

The interpretation is, for a 1-unit increase in X , the cumulative odds are multiplied by $\exp(\beta)$ for all cutpoints k .

2.2.3 Model selection

After fitting a proportional odds model with the number of affairs as the outcome and the presence of children, gender, religiousness, and self-rated marital satisfaction as predictors, a model containing all two-way interactions among these variables was explored. Backward selection based on AIC was then applied to identify the best-fitting model. The final selected model was a main-effects model that included children, religiousness, and marital satisfaction as predictors. The assumption of parallel line was carefully investigated for the model fit. This assumption was verified by a partial residual plot and a loglikelihood ratio test.

3. Missingness and Handling Methods

3.1 Missing Mechanism

In the simulation study, the MCAR mechanism was used for its simplicity. Under MCAR, the probability that a value is missing does not depend on either the observed or unobserved data.

3.2 Missing Data Approaches

To address missingness in the study, four missing-data approaches were used: CCA, EM-based single imputation, EM-based multiple imputation and MICE.

3.2.1 Complete Case Analysis

Complete case analysis operates on a dataset by using only subjects with fully observed data, removing any rows that contain missing values for the variables included in the analysis.

3.2.2 Expectation Maximization Algorithm

This subsection explains the algorithm used for both SIEM and EM. This imputation method was implemented via maximum-likelihood estimation using fractional augmentation in the E-step and weighted proportional odds regression in the M-step, yielding a single set of maximum likelihood estimates under a joint model for the outcome and predictors (Maity et al. 2025).

Consider $\theta = (\alpha, \beta)$ from the equation (1) and a joint model was assumed

$$f(Y, X; \theta, \pi) = f(Y|X; \theta) \prod_{j=1}^p f(X_j; \pi_j)$$

where $f(Y|X; \theta)$ is a proportional odds model and π_j denotes the marginal category probabilities of X_j .

Starting from complete-case estimates $(\theta^{(0)}, \pi^{(0)})$, parameters are updated by the EM algorithm.

In the E-step, for each observation with missing values, all possible realizations of the missing variable are enumerated and assigned posterior weights. For missing outcomes,

$$w_{ik} = \Pr(Y_i = k | X_i; \theta^{(t)}),$$

and for a missing predictor X_{ij}

$$w_{ijl} \propto \pi_{jl}^{(t)} \Pr(Y_i = k | X_i; \theta^{(t)}),$$

with normalization so that weights sum to one. Fully observed observations are retained with weight one.

In the M-step, parameters are updated by maximizing the weighted complete-data likelihood: θ is obtained via weighted proportional odds regression, and π_j are updated as weighted category proportions. The algorithm iterates until convergence, yielding a single set of maximum-likelihood estimates θ_{EM} .

3.2.3 Algorithm of imputation by chained equations

MICE uses a fully conditional specification (FCS) model (van Buuren 2018). Specifically, MICE specifies a separate conditional model for each variable with missing data and iterating through these models. Let p be the total number of variables in a dataset and let $j = 1, \dots, J$ index the variables containing missing values, where $J \leq p$. For a given variable Z_j with missing values, let X denote the set of other variables in the dataset that are used as predictors in the imputation model.

For an ordinal variable Z_j , MICE uses a proportional odds model

$$P(Z_j \leq k|X; \theta), k = 1, \dots, K - 1.$$

Within FCS algorithm, each Z_j is regressed on the current values of the other variables, and missing values in Z_j is imputed from the corresponding conditional distribution. MICE uses this algorithm and cycling over all incomplete variables yields one iteration.

3.2.4 Multiple imputations

Both Expectation Maximization based multiple imputation and MICE are implemented by using a common “Bayesian-like” scheme (Rubin, 1987; Schafer, 1997). Let m be the iteration indicator $t = 1, \dots, m$, and for each imputation methods, estimate model parameters by each method and obtain $\hat{\theta}^{(base)}$ and $\hat{Var}(\hat{\theta}^{(base)})$. Draw $\theta^{(t)}$ from

$$\theta^{(t)} \sim N(\hat{\theta}^{(base)}, \hat{Var}(\hat{\theta}^{(base)}))$$

then generate missing values from the predictive distribution

$$Z_{mis}^{(t)} \sim f(Z_{mis} | Z_{obs}, \theta^{(t)}).$$

After imputing the missing values, we fit proportional odds model for each imputed dataset and combine the m estimations using Rubin’s rule.

3.2.5 Rubin’s Rule

EM and MICE generate m imputed datasets, so the parameter estimates and its variance must be combined across imputations to obtain single estimation. To do so, Rubin’s rule is applied.

Let $t = 1, \dots, m$ index of imputed datasets, θ_t the estimate from dataset t , and U_t its corresponding estimated variance. The pooled estimate is the average of m estimates:

$$\theta = \frac{1}{m} \sum_{t=1}^m \theta_t.$$

The within imputation variance is the average of m estimated variances:

$$U = \frac{1}{m} \sum_{t=1}^m U_t.$$

The between imputation variance is:

$$B = \frac{1}{m-1} \sum_{t=1}^m (\theta_t - \theta)^2.$$

Total variance is given by:

$$T = U + (1 + \frac{1}{m}) B.$$

Inference such as confidence intervals are constructed by using θ and T .

4. Simulation Study

4.1 Missingness

For the simulation study, three missingness rates were considered: 5%, 15%, and 30%. For each missingness rate, $D = 500$ simulated datasets were generated by repeatedly applying the specified missingness mechanism to the original dataset, Affairs, but only to the variables selected in the model through backward selection. Each incomplete dataset was then analyzed using the four approaches aforementioned.

4.2 Evaluation Criteria

For each missingness and each method, the RMSE and the bias of the regression coefficients, and its standard errors (SEs) were used as the criteria.

Let D denote the number of missingness introduced datasets and θ a vector of reference parameters from the original dataset (no missingness). For dataset $d = 1, \dots, D$, let $\theta^{(m)}$ be the corresponding estimate obtained by a given method. The RMSE of θ is defined as

$$RMSE = \sqrt{\frac{1}{D} \sum_{d=1}^D (\theta^{(m)} - \theta)^2}.$$

In this setting, RMSE was computed separately for each regression coefficient and the estimated SE of these coefficients. The bias of θ is defined as $\bar{\theta} - \theta_{reference}$, where $\bar{\theta}$ is the mean of θ across all repetitions D .

5. Results

Table1. Bias of Coefficients

Terms	5% missingness				15% missingness				30% missingness			
	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE
Intercept 1	0.0029	0.0015	0.0047	0.0003	0.0086	0.0082	0.0171	-0.0017	0.0038	0.0128	0.0351	-0.0054
Intercept 2	0.0043	0.0030	0.0038	0.0018	0.0122	0.0109	0.0129	0.0006	0.0159	0.0158	0.0242	-0.0036
childrenyes	-0.0033	-0.0025	-0.0015	-0.0001	-0.0061	-0.0095	-0.0074	0.0029	0.0024	-0.0154	-0.0159	0.0096
religiousness.L	0.0017	0.0029	-0.0195	0.0021	0.0002	0.0003	-0.0683	-0.0040	0.0227	0.0111	-0.1293	0.0045
religiousness.Q	0.0044	0.0027	0.0010	0.0056	-0.0017	0.0034	-0.0040	0.0097	-0.0140	0.0112	-0.0028	0.0256
religiousness.C	-0.0041	-0.0004	0.0003	-0.0019	-0.0021	-0.0039	-0.0017	-0.0080	0.0086	0.0001	0.0032	-0.0063
religiousness^4	-0.0016	0.0009	0.0004	0.0078	0.0062	0.0032	0.0015	0.0251	-0.0090	-0.0030	-0.0021	0.0454
rating.L	-0.0024	0.0014	-0.0190	-0.0056	-0.0039	0.0031	-0.0537	-0.0107	0.0010	0.0102	-0.1088	-0.0245
rating.Q	-0.0031	-0.0021	-0.0052	-0.0012	0.0021	0.0005	-0.0116	0.0009	0.0066	-0.0012	-0.0280	0.0008
rating.C	0.0027	0.0013	0.0130	0.0030	-0.0042	-0.0019	0.0309	0.0070	-0.0077	0.0012	0.0700	0.0210
rating^4	0.0022	0.0016	-0.0026	-0.0031	0.0044	0.0043	-0.0069	-0.0073	-0.0001	0.0061	-0.0188	-0.0226

* Single Imputation with EM

** Multiple Imputation with EM

At 5% missingness, all methods exhibit very small bias across all coefficients. MICE most frequently achieve the lowest bias, while SIEM also performs well and is often competitive with MICE. At 15% missingness, CCA, SIEM, and MICE perform relatively similarly on average in terms of bias, whereas EM performs substantially worse, exhibiting the highest bias across several coefficient categories. The same pattern persists at 30% missingness, with CCA, SIEM, and MICE remaining comparable, and EM again showing noticeably larger bias, mirroring the trend observed at 15%.

Table2. Bias of Standard Error

Terms	5% missingness				15% missingness				30% missingness			
	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE
Intercept 1	0.0067	-0.0005	0.0027	0.0025	0.0221	-0.0011	0.0091	0.0082	0.0523	-0.0020	0.0175	0.0166
Intercept 2	0.0071	-0.0005	0.0028	0.0025	0.0232	-0.0009	0.0093	0.0083	0.0547	-0.0017	0.0180	0.0165
childrenyes	0.0066	-0.0006	0.0030	0.0026	0.0222	-0.0012	0.0098	0.0089	0.0519	-0.0023	0.0194	0.0179
religiousness.L	0.0083	0.0003	0.0040	0.0044	0.0256	0.0002	0.0109	0.0118	0.0604	0.0009	0.0223	0.0243
religiousness.Q	0.0074	0.0002	0.0040	0.0039	0.0233	0.0003	0.0108	0.0107	0.0548	0.0008	0.0208	0.0211
religiousness.C	0.0060	0.0000	0.0030	0.0030	0.0193	0.0001	0.0086	0.0091	0.0456	0.0005	0.0178	0.0173
religiousness^4	0.0052	0.0000	0.0025	0.0025	0.0172	0.0003	0.0082	0.0077	0.0399	0.0005	0.0165	0.0152
rating.L	0.0095	-0.0001	0.0045	0.0043	0.0327	0.0012	0.0153	0.0147	0.0785	0.0028	0.0314	0.0299
rating.Q	0.0086	0.0000	0.0044	0.0041	0.0297	0.0012	0.0143	0.0134	0.0707	0.0025	0.0294	0.0269
rating.C	0.0068	0.0000	0.0036	0.0035	0.0225	0.0008	0.0109	0.0106	0.0532	0.0017	0.0223	0.0215
rating^4	0.0061	0.0000	0.0032	0.0031	0.0200	0.0002	0.0100	0.0089	0.0464	0.0005	0.0218	0.0182

* Single Imputation with EM

** Multiple Imputation with EM

Across all missingness percentages, a consistent pattern emerges. CCA performs the worst in terms of bias, followed by MICE and EM, which perform similarly, while SIEM consistently achieves the lowest bias. In addition, all methods tend to overestimate the standard errors, although SIEM occasionally underestimates the standard errors.

Table3. RMSE of Coefficients

Terms	5% missingness				15% missingness				30% missingness			
	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE
Intercept 1	0.0559	0.0396	0.0408	0.0386	0.1004	0.0715	0.0748	0.0639	0.1630	0.1088	0.1139	0.1013
Intercept 2	0.0589	0.0412	0.0423	0.0402	0.1080	0.0744	0.0764	0.0669	0.1737	0.1136	0.1151	0.1056
childrenyes	0.0555	0.0423	0.0435	0.0416	0.1048	0.0765	0.0783	0.0705	0.1631	0.1179	0.1194	0.1125
religiousness.L	0.0682	0.0519	0.0570	0.0527	0.1289	0.0844	0.1069	0.0873	0.1826	0.1256	0.1750	0.1293
religiousness.Q	0.0640	0.0434	0.0440	0.0450	0.1104	0.0781	0.0739	0.0779	0.1737	0.1107	0.0982	0.1100
religiousness.C	0.0505	0.0348	0.0353	0.0354	0.0922	0.0655	0.0629	0.0649	0.1530	0.0984	0.0865	0.0915
religiousness^4	0.0447	0.0324	0.0331	0.0339	0.0860	0.0574	0.0563	0.0627	0.1226	0.0818	0.0780	0.0904
rating.L	0.0847	0.0571	0.0612	0.0576	0.1568	0.1091	0.1195	0.1031	0.2641	0.1691	0.1930	0.1536
rating.Q	0.0755	0.0544	0.0549	0.0539	0.1415	0.0959	0.0936	0.0923	0.2353	0.1479	0.1385	0.1315
rating.C	0.0603	0.0442	0.0457	0.0436	0.1079	0.0750	0.0775	0.0713	0.1818	0.1157	0.1260	0.1070
rating^4	0.0533	0.0380	0.0390	0.0382	0.0968	0.0654	0.0654	0.0643	0.1492	0.0955	0.0977	0.0931

*Single Imputation with EM

**Multiple Imputation with EM

For the RMSE of the coefficients, the same pattern is observed across all missingness percentages. CCA performs substantially worse than the other methods at all levels of missingness. In contrast, SIEM, EM, and MICE perform relatively similar, with comparable RMSE values across most coefficients and missingness levels.

Table4. RMSE of Standard Error

Terms	5% missingness				15% missingness				30% missingness			
	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE	CCA	SIEM*	EM**	MICE
Intercept 1	0.0079	0.0025	0.0041	0.0037	0.0236	0.0044	0.0110	0.0099	0.0542	0.0065	0.0202	0.0190
Intercept 2	0.0082	0.0025	0.0042	0.0037	0.0247	0.0045	0.0113	0.0101	0.0565	0.0067	0.0205	0.0190
childrenyes	0.0075	0.0023	0.0042	0.0037	0.0233	0.0042	0.0115	0.0105	0.0533	0.0062	0.0216	0.0203
religiousness.L	0.0095	0.0027	0.0054	0.0056	0.0271	0.0046	0.0128	0.0137	0.0621	0.0066	0.0250	0.0266
religiousness.Q	0.0082	0.0020	0.0050	0.0048	0.0243	0.0033	0.0121	0.0120	0.0559	0.0050	0.0229	0.0230
religiousness.C	0.0063	0.0010	0.0034	0.0035	0.0197	0.0019	0.0094	0.0099	0.0461	0.0028	0.0192	0.0185
religiousness^4	0.0054	0.0008	0.0029	0.0028	0.0174	0.0015	0.0088	0.0083	0.0402	0.0020	0.0175	0.0162
rating.L	0.0133	0.0040	0.0075	0.0071	0.0382	0.0084	0.0203	0.0191	0.0856	0.0123	0.0374	0.0352
rating.Q	0.0113	0.0031	0.0065	0.0062	0.0335	0.0066	0.0178	0.0170	0.0756	0.0097	0.0340	0.0306
rating.C	0.0076	0.0015	0.0044	0.0044	0.0237	0.0031	0.0123	0.0120	0.0547	0.0046	0.0242	0.0232
rating^4	0.0064	0.0011	0.0038	0.0036	0.0204	0.0020	0.0109	0.0098	0.0470	0.0031	0.0232	0.0194

*Single Imputation with EM

**Multiple Imputation with EM

For the RMSE of the standard errors, the same pattern holds across all missingness percentages. CCA performs by far the worst at all levels of missingness. EM and MICE perform similarly to one another and consistently outperform CCA across all coefficients. However, SIEM performs the best overall, consistently achieving the lowest RMSE for the standard errors among all four methods.

6. Conclusion

In general, CCA performs the worst in terms of bias of the standard errors and RMSE for both the coefficients and the standard errors; however, it performs relatively well compared to the other methods with respect to coefficient bias. MICE and EM exhibit broadly similar performance across all four criteria, typically outperforming CCA but underperforming relative to SIEM, with the exception of coefficient bias, where EM performs poorly at 15% and 30% missingness. SIEM performs consistently well across all four criteria and clearly

outperforms the other methods in terms of bias and RMSE of the standard errors, indicating strong ability to accurately replicate the reference model variability. Overall, when considering all evaluation criteria jointly, SIEM emerges as the best-performing method, followed by MICE and EM as close second-tier approaches, while CCA performs comparatively poorly relative to the other three methods.

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